

Evenness Score — Unsupervised Per-Step Metric

a 0–100 gait-regularity index computed without surface labels

WalkSensePlace

The classification models predict a surface *label*. The **evenness score** is different: it assigns each step a continuous value from **0 (very uneven / rough)** to **100 (very even / smooth)** using **no surface labels anywhere**. It is an unsupervised *scoring model* — a robust scale is fitted on a set of recordings, after which any new step can be scored on the same 0–100 axis. It measures how regular the gait is, within a step and from step to step; surface tags, where they exist, are used only to *validate* the score, never to build it.

Notation

For one step (gait cycle) of length n samples at $f_s = 62.5$ Hz: total pressure p_t ; the 12 pad values $s_{i,t}$ ($i = 1, \dots, 12$); accelerometer a_t^x, a_t^y, a_t^z with magnitude $\|a_t\|$; gyroscope g_t^x, g_t^y, g_t^z (median-centred within the step). The mean over the step is $\bar{p} = \frac{1}{n} \sum_t p_t$. The *flat-foot instant* $t^* = \arg \min_t \|g_t\|$ is where the foot is momentarily still and the accelerometer points along gravity, giving two tilt axes

$$\theta_A = \text{atan2}(a^x, \sqrt{(a^y)^2 + (a^z)^2}), \quad \theta_B = \text{atan2}(a^z, \sqrt{(a^x)^2 + (a^y)^2}) \quad (\text{degrees}).$$

Steps are detected from peaks of p_t ; windows shorter than 15 or longer than 150 samples are skipped. No surface markers are read at any point.

1 The irregularity features

Every feature is oriented so that **larger** = **more uneven**. There are two groups.

1.1 Within-step irregularity (how unsteady a single step is)

- **orient_range_a**, **orient_range_b** — the range of foot tilt *during* the step (how much the foot rocks while planted): $\max_t \theta_{A,t} - \min_t \theta_{A,t}$ and likewise for θ_B .
- **cop_wander** — how much the pressure distribution shifts around the sole frame to frame. With the per-frame normalised map $Q_{i,t} = s_{i,t} / \sum_j s_{j,t}$,

$$\text{cop_wander} = \frac{1}{n-1} \sum_{t=2}^n \sum_{i=1}^{12} |Q_{i,t} - Q_{i,t-1}|. \quad (1)$$

- **press_jitter** — small pressure fluctuations, as the mean over pads of the standard deviation of the per-pad second difference, session-normalised:

$$\text{press_jitter} = \frac{1}{12\bar{p}} \sum_{i=1}^{12} \text{std}_t(\Delta^2 s_{i,t}), \quad \Delta^2 s_{i,t} = s_{i,t} - 2s_{i,t-1} + s_{i,t-2}. \quad (2)$$

- **gyro_std** — rotational wobble, the mean over axes of the angular-rate standard deviation: $\frac{1}{3} \sum_{c \in \{x,y,z\}} \text{std}_t(g_t^c)$.

1.2 Step-to-step consistency (how much a step differs from its neighbours)

For each of the following per-step scalars we take the **rolling 3-step standard deviation** within a recording+foot sequence; for a per-step scalar $u^{(k)}$ (step index k),

$$\text{roll3_std}(u)^{(k)} = \sqrt{\frac{1}{3} \sum_{j=k-2}^k (u^{(j)} - \bar{u}^{(k)})^2}, \quad \bar{u}^{(k)} = \frac{1}{3} \sum_{j=k-2}^k u^{(j)}. \quad (3)$$

The scalars are:

- **stance_dur** — loaded fraction of the step, $\frac{1}{n} \#\{t : p_t > \frac{1}{2}\bar{p}\}$.
- **loading_rate** — normalised peak rise rate, $\max_t(\Delta p_t)/\bar{p}$.
- **impact** — $\max_{t \leq n/3} \|a_t\| / \text{median}_t \|a_t\|$.
- **tiltA, tiltB** — the flat-foot tilt θ_A, θ_B at t^* .

Applying Eq. (??) gives **stance_dur_roll3_std**, **loading_rate_roll3_std**, **impact_roll3_std**, **tiltA_roll3_std**, **tiltB_roll3_std**. A high value means the quantity jumps around from step to step — the hallmark of walking on irregular ground.

The full set is the $F = 10$ features **{orient_range_a, orient_range_b, cop_wander, press_jitter, gyro_std}** (within-step) plus the five ***_roll3_std** (step-to-step).

2 From features to a 0–100 score

Robust standardisation. Each feature is standardised by its *median* and *interquartile range* ($\text{IQR} = Q_{75} - Q_{25}$), which resist outliers far better than mean/std, and clipped so no single step dominates:

$$z_f = \text{clip}\left(\frac{x_f - m_f}{s_f}, -4, 4\right), \quad m_f = \text{median}(x_f), \quad s_f = \text{IQR}(x_f). \quad (4)$$

Unevenness index. The standardised features are averaged with equal weight into a single *unevenness index* (higher = rougher):

$$U = \frac{1}{F} \sum_{f=1}^F z_f. \quad (5)$$

Equal weighting keeps the score transparent — any step’s value can be traced back to which features pushed it up or down — and, because every feature is an independent irregularity measure pointing the same way, they reinforce rather than cancel.

Calibration to 0–100. U is itself robustly standardised and passed through the standard-normal CDF Φ , then scaled, so the result is bounded, calibrated, and transferable:

$$\text{evenness} = 100 \cdot \Phi\left(-\frac{U - m_U}{s_U}\right), \text{ clipped to } [0, 100]. \quad (6)$$

The minus sign flips rough→smooth so that **higher** = **more even**. A step at the median unevenness scores 50; a step two (robust) standard deviations smoother scores ≈ 98 , and one two deviations rougher ≈ 2 . So 50 is “a typical step,” and the score reads directly as a percentile-like position on the smoothness axis.

3 The fitted model

Fitting stores only the calibration constants:

$$\{ m_f, s_f \text{ for each feature } \} \cup \{ m_U, s_U \},$$

saved as `evenness_scaler.json`. Scoring a *new* recording re-uses these exact constants: extract the 10 features per step, apply the stored m_f, s_f , average to U , apply the stored m_U, s_U and Φ . Because the reference numbers are fixed at fit time, scores are comparable across recordings and walks — “50” always means the same thing (the median step of the fitting set), not “the median step of whatever file this happens to be.”

4 What the score measures (validation)

Although no labels are used to build it, grouping the score by surface (where tags exist) shows it lands exactly where physical intuition says it should:

surface	mean evenness	interpretation
grass	≈ 40	compliant, irregular — <i>least even</i>
loose	≈ 51 –55	firm, fairly regular
paved	≈ 51 –52	firm, regular

Two things follow. First, the score **recovers the compliance axis unsupervised**: grass (soft, uneven) is cleanly separated below the firm surfaces, with no target ever supplied. Second, it is **not** a paved-vs-loose discriminator — those score almost identically, which is honest: a smooth compact-dirt path really is about as regular to walk on as pavement, so a *gait-regularity* metric should not, and does not, tell them apart. Evenness measures how steady the walking is, which is the soft-vs-firm distinction, not surface material *per se*.

One-line reading

Evenness is an equal-weight, robustly-standardised composite of ten within-step and step-to-step irregularity features, calibrated through the normal CDF to a transferable 0–100 scale (50 = typical step, higher = smoother). With no labels, it reproduces the compliance axis — grass scores distinctly low, firm surfaces high — making it a label-free companion to the supervised soft-vs-hard model rather than a replacement for it.